# Political Affiliation Detection and Semantic Mapping of the French Archelec Corpus (1973–1993)

Rafaël Mourouvin
ML FOR NLP; ENSAE Paris
rafael.mourouvin@polytechnique.edu

April 30, 2026

**Abstract**

In this report, I study the Archelec corpus of French legislative electoral manifestos (1973 to 1993) through three complementary research questions. **RQ1** asks whether the political affiliation of a candidate can be predicted automatically from manifesto text alone. **RQ2** investigates whether politically close parties share a similar rhetorical space and whether the French political landscape can be mapped from text. **RQ3** evaluates whether a classifier trained on manifestos from 1973 to 1981 generalises to the 1988 and 1993 elections. I compare five models ranging from Naive Bayes to FastText embeddings and apply Latent Dirichlet Allocation (LDA) for unsupervised semantic mapping. The best model (TF-IDF + Linear SVM) achieves F1 = 0.963 on a random split after systematic removal of party-name leakage terms. The LDA semantic map reproduces the classical left/right structuring of French politics without any supervision on party positions, and the parties closest in topic space correspond exactly to the main confusion pairs of the classifier, providing a coherent joint validation of RQ1 and RQ2 from two fully independent methods. Temporal generalisation degrades to F1 = 0.507 when training on 1973 to 1981 and testing on 1988 and 1993, and leave-one-year-out F1 ranges from 0.528 (1993, peak disruption) to 0.793 (1981, peak bipolarisation), suggesting that classifier performance on held-out years tracks historical political discontinuity.

## 1 Introduction

Electoral manifestos are primary political documents that encode a party's programmatic commitments, rhetorical identity, and ideological positioning. Unlike party platforms or leader speeches, professions de foi are produced at the candidate level, giving a uniquely granular view of how political affiliation is expressed in written french across decades.

The Archelec corpus, assembled by Sciences Po and the CEVIPOF archelec, digitises and transcribes professions de foi submitted by candidates to the French National Assembly across five elections: 1973, 1978, 1981, 1988, and 1993. This twenty-year window covers deep structural changes in french politics: the consolidation of the RPR/UDF right-wing coalition, the rise of the Front National from a marginal force (around 100 candidates in 1973) to a mass movement (over 1,000 in 1993), the electoral decline of the left after 1981, and the emergence of ecology as an organised political force.

Automatic classification of political texts has a long tradition in political science and NLP, from bag-of-words approaches to neural representations. Most existing work focuses on english-language party manifestos at the national level. The Archelec corpus offers a unique french-language dataset with candidate-level granularity and strong temporal coverage.

This report answer three questions:

**RQ1: Classification:** Can we automatically predict the political affiliation of a candidate from manifesto text alone, and how do different text representations compare? I evaluate five models covering the full NLP spectrum from Naive Bayes to FastText embeddings, with a careful leakage-removal protocol to ensure that performance reflects genuine rhetorical differences (Section 3).

**RQ2: Semantic structure:** Do politically close parties share a similar rhetorical space? Can the French political landscape be mapped from manifesto text alone? I apply LDA topic modelling and build a semantic map using interpretable topic axes, then cross-validate this map against the confusion pairs of the classifier (Section 4).

**RQ3: Temporal stability:** Does political vocabulary remain stable enough across decades to generalise a classifier trained on 1973-1981 manifestos to 1988 and 1993? I evaluate this through a direct temporal split and a leave-one-year-out protocol, and interpret the results in light of historical political changes (Section 5).

# 2 Data

## 2.1 The Archelec Corpus

The corpus contains 21,167 matched documents after merging OCR-transcribed text files (Arkindex platform) with candidate metadata downloaded from `archelec.sciencespo.fr`. Each document is a profession de foi associated with a candidate, an election year, a department, a constituency, and a political affiliation label (`titulaire-soutien`). The transcription was produced by automated OCR on scanned originals, and noise is present but generally limited to isolated character errors.

## 2.2 Label Engineering

The raw `titulaire-soutien` field contains 1,523 distinct strings, reflecting inconsistent data entry across decades and the fragmented nature of French political coalitions. I apply three principled cleaning steps before any modelling.

**Step 1: Remove missing labels.** 4,040 documents (19%) carry the label "non mentionné". This reflects missing metadata rather than a real political category, and including these documents would inject noise into any classifier. They are excluded from all supervised analyses.

**Step 2: Merge label variants.** The same political grouping systematically appears under multiple orderings. I consolidate five RPR/UDF variants (e.g. "RPR;UDF" and "UDF;RPR") into a single `RPR / UDF` label, merge "Parti socialiste + MRG" into `Parti socialiste`, and unify three Verts/Génération Écologie variants into `Écologistes`. These merges reflect political reality: the RPR/UDF coalition presented joint candidates but the labelling order was arbitrary, while the PS + MRG alliance had a unified programme.

**Step 3: Frequency threshold.** I retain only parties with at least 120 documents to ensure reliable classification, yielding 10 classes and 11,290 documents (Table 1).

## 2.3 Corpus Statistics

Figure 1 shows the per-party per-year manifesto counts and reveals two historically important artefacts. First, the Mouvement réformateur appears only in 1973: it was an ephemeral centrist coalition that dissolved into the UDF in 1978. As a single-year class, it is trivially learnable on random splits but entirely absent from temporal test sets, which is a relevant caveat for RQ3. Second, the Écologistes appears only in 1993: the Verts were founded in 1984 and only fielded a mass candidacy (795 documents) in 1993 alongside Génération Écologie.

Table 1: Class distribution after label engineering.

| Party | Documents | Years present |
|---|---|---|
| Parti communiste français | 2,474 | 1973,1993 |
| RPR / UDF | 2,452 | 1973,1993 |
| Parti socialiste | 2,182 | 1973,1993 |
| Front national | 1,366 | 1973,1993* |
| Lutte ouvrière | 1,023 | 1973,1993 |
| Écologistes | 795 | 1993 only |
| Parti socialiste unifié | 441 | 1973,1988 |
| Ligue communiste révolutionnaire | 219 | 1973,1993 |
| Mouvement réformateur | 205 | 1973 only |
| indépendant | 133 | 1973,1993 |
| **Total** | **11,290** | |

*FN marginal pre-1988: 92, 41, 44, 543, 646 docs per year.

Figure 2 shows the document length distribution. The median is 677 tokens (mean 738, std 356), with a long right tail from candidates who submitted extended multi-page manifestos. Left-wing radical parties (Lutte ouvrière, LCR, PSU) produce systematically longer documents (median 900,1100 tokens) than centrist and right-wing ones (RPR/UDF, Écologistes, median 400,500 tokens), which is consistent with a more programmatic writing style on the radical left.

### 2.4 Preprocessing

**Leakage removal.** A critical design issue for this task is label leakage: party names appearing directly in manifesto text trivially reveal the label. For instance, the word ”communiste” almost always signals PCF affiliation, and ”Front national” directly signals FN. Without removal, a classifier could achieve high accuracy simply by detecting party names rather than learning genuine rhetorical patterns, which would make the results uninterpretable. I remove 18 leakage terms (party names and abbreviations) from all documents before any model training, ensuring that discriminative features reflect genuine rhetorical and ideological differences.

**Language filtering (LDA only).** Candidates in Alsace-Moselle wrote partly in German or Alsatian. In a preliminary LDA run, this produced a spurious topic entirely dominated by German function words (”die”, ”der”, ”und”, ”ist”, ”den”), which carries no political information and contaminates the thematic structure of all other topics. I detect and exclude 323 such documents using a simple heuristic (threshold: at least 3 German marker words in the document vocabulary) for the LDA analysis only, retaining them for classification. The LDA corpus is thus reduced to 10,967 French documents.

## 3 RQ1: Political Affiliation Classification

### 3.1 Models and Experimental Setup

I evaluate five models following the progression from classic bag-of-words to dense neural representations:

**Naive Bayes (baseline).** Multinomial NB with TF-IDF features (unigrams and bigrams, 20,000 features, `sublinear_tf=False`). Naive Bayes is included as a lower-bound baseline because its conditional independence assumption is particularly problematic for bigrams, where word co-occurrences carry important

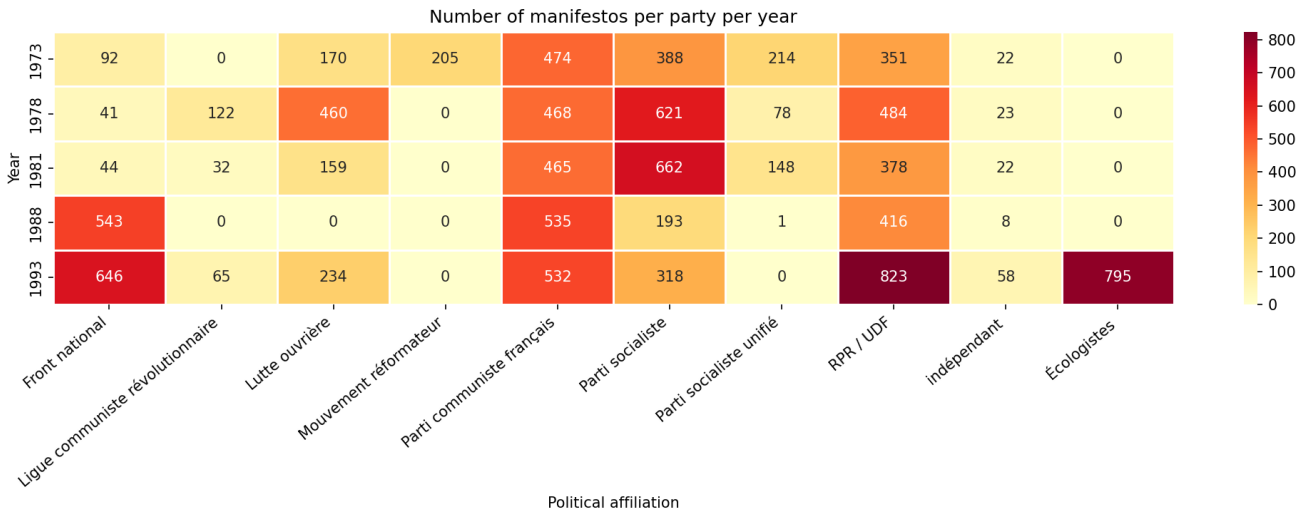


Figure 1: Number of manifestos per party per year. The FN grows from 92 (1973) to 646 (1993), multiplying by 6.7 between 1981 and 1988. Écologistes appear only in 1993; Mouvement réformateur only in 1973.

political meaning (e.g. "programme commun" vs. the individual words in isolation). The absence of log normalisation also amplifies high-frequency generic terms at the expense of rare but distinctive ones.

**TF-IDF + Logistic Regression.** Log-normalised TF-IDF (20,000 features, `sublinear_tf=True`, unigrams and bigrams) with L2-regularised LR (`class_weight=balanced`, $C = 1$, `max_iter=1000`). The balanced weighting corrects for class imbalance between the large parties (PCF, 2,474 documents) and the small ones (indépendant, 133 documents).

**TF-IDF + Linear SVM.** Same TF-IDF vectorisation with LinearSVC (`class_weight=balanced`, $C = 1$, `max_iter=2000`). Linear SVMs are known to perform well on high-dimensional sparse text features, where the large margin criterion is particularly effective.

**TF-IDF + LSI + Logistic Regression.** Truncated SVD ($k = 200$ components, explaining 39.3% of variance) applied to the TF-IDF matrix, producing dense low-dimensional document representations that capture latent semantic structure. LSI sits between sparse TF-IDF and neural embeddings in terms of representation.

**FastText + Logistic Regression.** A FastText skip-gram model bojanowski2017 (dim=100, window=5, 10 epochs, min_count=3) trained from scratch on the training corpus. Document vectors are obtained by mean-pooling word vectors, then classified by LR. Unlike TF-IDF, FastText uses character n-gram subword representations, which can help with morphological variants in French political vocabulary.

All models use a stratified 80/20 train-test split (`random_state=42`, 9,032 training and 2,258 test documents). I report macro-averaged F1 as the primary metric, giving equal weight to all classes regardless of size.

### 3.2 Results and Analysis

Figure 3 summarises the comparison. TF-IDF + SVM achieves the best overall performance (F1 = 0.963, accuracy = 0.978). The strong advantage of sparse TF-IDF over dense representations can be explained by the nature of political text: party rhetoric relies on distinctive, repetitive vocabulary, specific terms like "travailleurs", "immigration", or "autogestion" that appear frequently in one party's manifestos but rarely in others. TF-IDF weights naturally amplify exactly this kind of signal. LSI (F1 = 0.905) sacrifices some of this discriminative sparsity by compressing 20,000 features into 200 dense dimensions. FastText (F1 = 0.818)

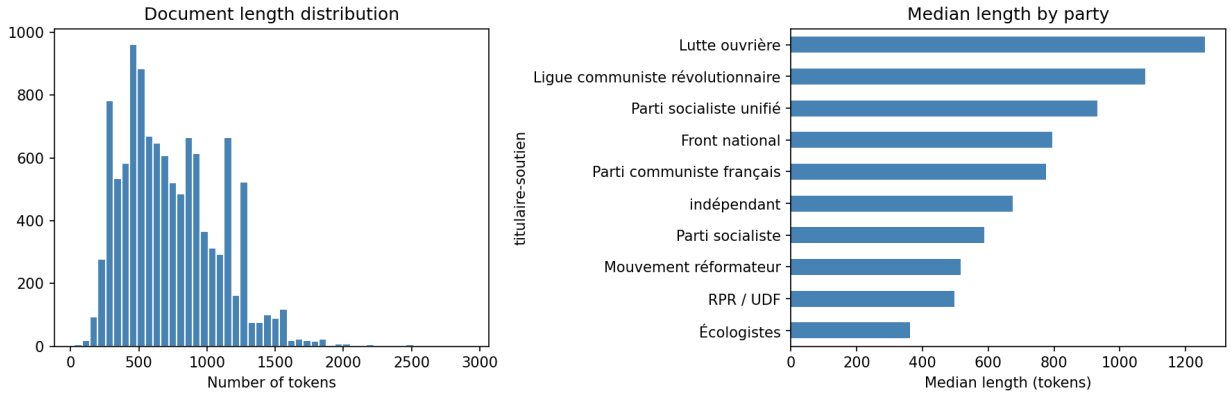


Figure 2: Document length distribution (left) and median length by party (right). Left-wing radical parties write substantially longer manifestos.

Table 2: Classification results on the random 80/20 test split (leakage-free). Best result in bold.

| **Model** | **Representation** | **Accuracy** | **F1 macro** |
|---|---|---|---|
| Naive Bayes | BoW counts (sparse) | 0.912 | 0.779 |
| TF-IDF + LR | TF-IDF sparse | 0.973 | 0.950 |
| **TF-IDF + SVM** | **TF-IDF sparse** | **0.978** | **0.963** |
| TF-IDF + LSI + LR | TF-IDF + SVD dense | 0.943 | 0.905 |
| FastText + LR | FastText dense | 0.860 | 0.818 |

underperforms the TF-IDF models despite its morphological awareness, because the embeddings are trained on a small corpus of around 9,000 documents. Pre-trained French embeddings such as `cc.fr.300` or CamemBERT martin2020 would likely close this gap. Naive Bayes (F1 = 0.779) lags far behind, largely because its conditional independence assumption is violated by the politically meaningful bigrams.

The confusion matrix of the best model (Figure 4) shows that nearly all parties achieve F1 $\geq$ 0.95. Two exceptions stand out. First, indépendant (F1 = 0.81): independent candidates have no party vocabulary to draw on, and their manifestos cover a wide variety of rhetorical styles, which makes the class hard to learn. Second, the residual confusion between PS and RPR/UDF (11 errors) and between PCF and PS (8 errors) suggests that some centrist PS candidates and some moderate RPR/UDF candidates share rhetorical common ground. I analyse these confusions in depth in Section 4.4.

Figure 5 shows the top discriminative words per party after leakage removal. The features are interpretable and politically meaningful: the FN is discriminated by ”immigration” and ”français d’abord”; LO by ”travailleuses” and ”arlette laguiller”; the PCF by ”gauche” and ”forces progressistes”; RPR/UDF by ”union candidate” and ”nouvelle majorité”. This validates both the leakage removal (no party names appear as top features) and the underlying hypothesis that parties have genuine lexical signatures that go beyond simply naming themselves.

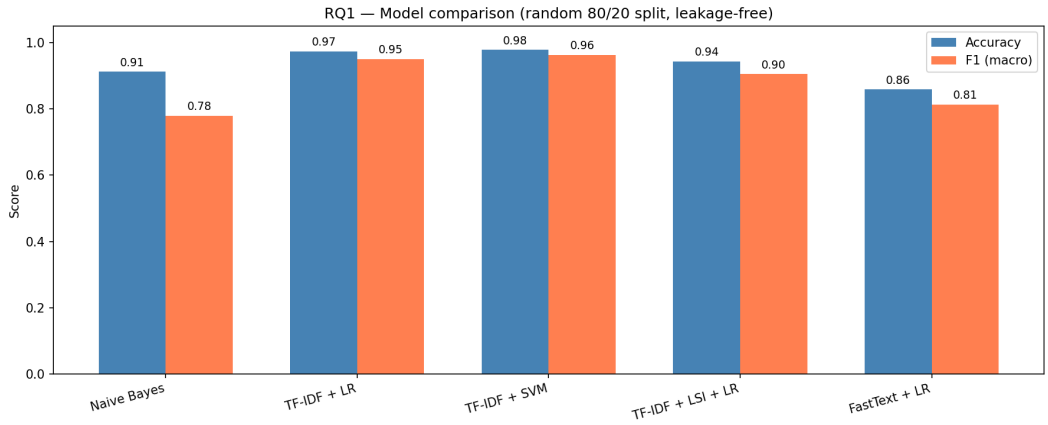


Figure 3: RQ1: model comparison on the random 80/20 split (leakage-free). TF-IDF + SVM dominates all models on both metrics.

# 4 RQ2: Semantic Mapping of the Political Landscape

## 4.1 Motivation and Approach

RQ1 showed that political parties have distinctive lexical signatures. RQ2 asks a complementary question: is there a semantic structure to these signatures? Do parties that the classifier confuses share similar thematic patterns? Can I map the French political landscape from manifesto text alone, without any supervision on political positions?

I address this through LDA topic modelling, an entirely unsupervised method that operates on word co-occurrence patterns without any knowledge of party labels. The key methodological strength is independence: LDA and TF-IDF-SVM are trained on completely different objectives. Any convergence between their outputs, that is, any alignment between topic proximity and classification confusion, cannot be a methodological artefact and must reflect genuine political structure in the text.

## 4.2 LDA Topic Modelling

I apply LDA ($K = 8$ topics, 20 online EM iterations, `random_state=42`) to the 10,967 French documents (Section 2.4). I chose $K = 8$ based on the number of meaningful political cleavages visible in the corpus: class struggle, nationalism, ecology, institutional discourse, and the general left/right axis. Figure 6 shows the top words per topic, and Table 3 gives the labels I assigned after inspection.

The topic-party heatmap (Figure 7) shows how each party's documents distribute across topics, and several patterns stand out. LCR concentrates almost entirely on Topic 2 (Labour and class struggle, 97%), reflecting its Trotskyist orientation. RPR/UDF and indépendant are dominated by Topic 4 (Electoral programme, 100%), a general register that avoids ideological specificity, consistent with a coalition that needed to appeal broadly. Écologistes split predictably between Topic 5 (Ecology, 49%) and Topic 4 (Electoral programme, 44%). Front national splits between Topic 0 (Nationalist and immigration, 43%) and Topic 6 (Nationalist electoral, 46%), capturing its dual identity as both an identity-politics movement and an electoral challenger. PCF and PS both draw heavily on Topics 1,3 but in different proportions, which explains why they are rhetorical neighbours yet lexically distinct.

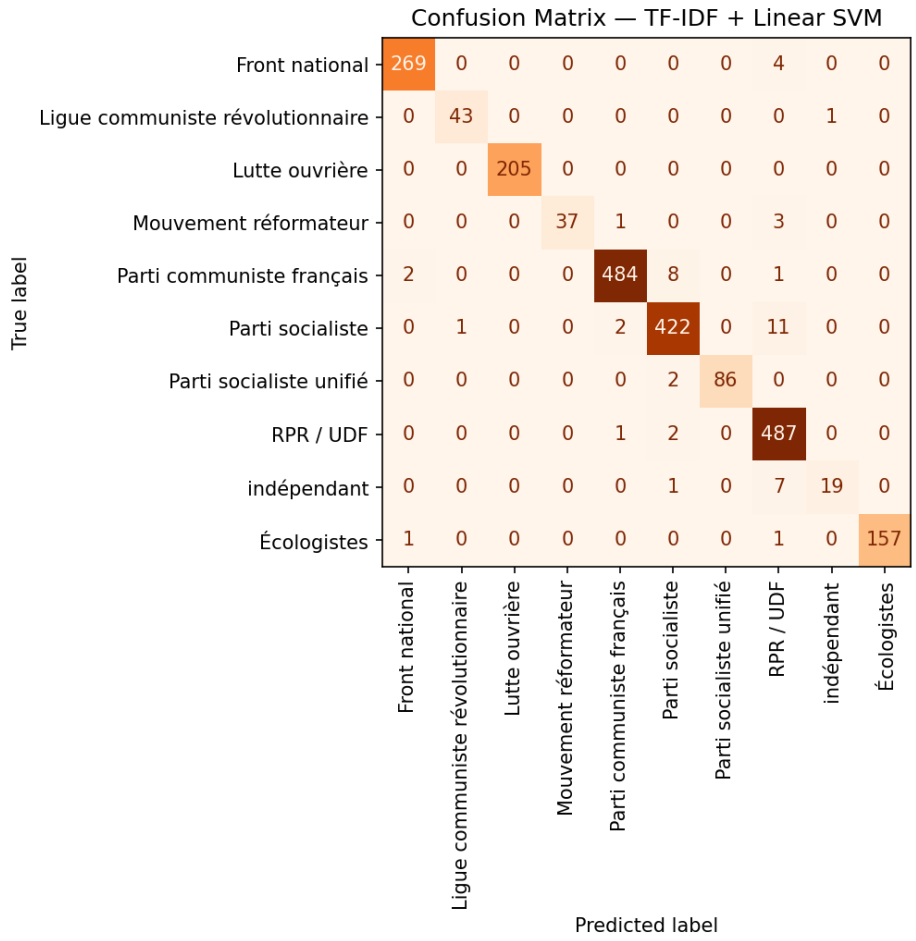


Figure 4: Confusion matrix of the best model (TF-IDF + Linear SVM). Errors are concentrated between rhetorically proximate parties.

### 4.3 Semantic Map of the Political Landscape

For each party, I compute the centroid of its document vectors in the 8-dimensional LDA topic space. I then select the two topics with the highest variance across party centroids as the axes of a 2D map: Topic 4 (Electoral programme, x-axis) and Topic 1 (Labour and left/right discourse, y-axis). These are the two dimensions that maximally separate the ten parties in topic space, and crucially, each axis corresponds to a named, substantively meaningful thematic dimension. This is more interpretable than a t-SNE projection, where axes have no direct meaning.

Figure 8 shows the resulting map. The spatial structure reproduces the classical left-right axis of French politics. Left-wing radical parties (PCF, LCR, PSU) cluster in the upper-left quadrant, with high Labour and left/right scores and low Electoral programme scores. RPR/UDF occupies the lower-right corner with high Electoral programme and low Labour scores. The Parti socialiste lies in the middle right, consistent with its social-democratic positioning between the communist left and the moderate right. Front national is isolated in the middle: its nationalist discourse (Topics 0 and 6) does not project strongly onto either axis, confirming its transversal position in the French political landscape. Lutte ouvrière sits at the far upper-left with a near-zero Electoral programme score, reflecting its principled rejection of electoral politics as an end. Écologistes sits at the centre bottom, dominated by neither the left/right axis nor the electoral programme

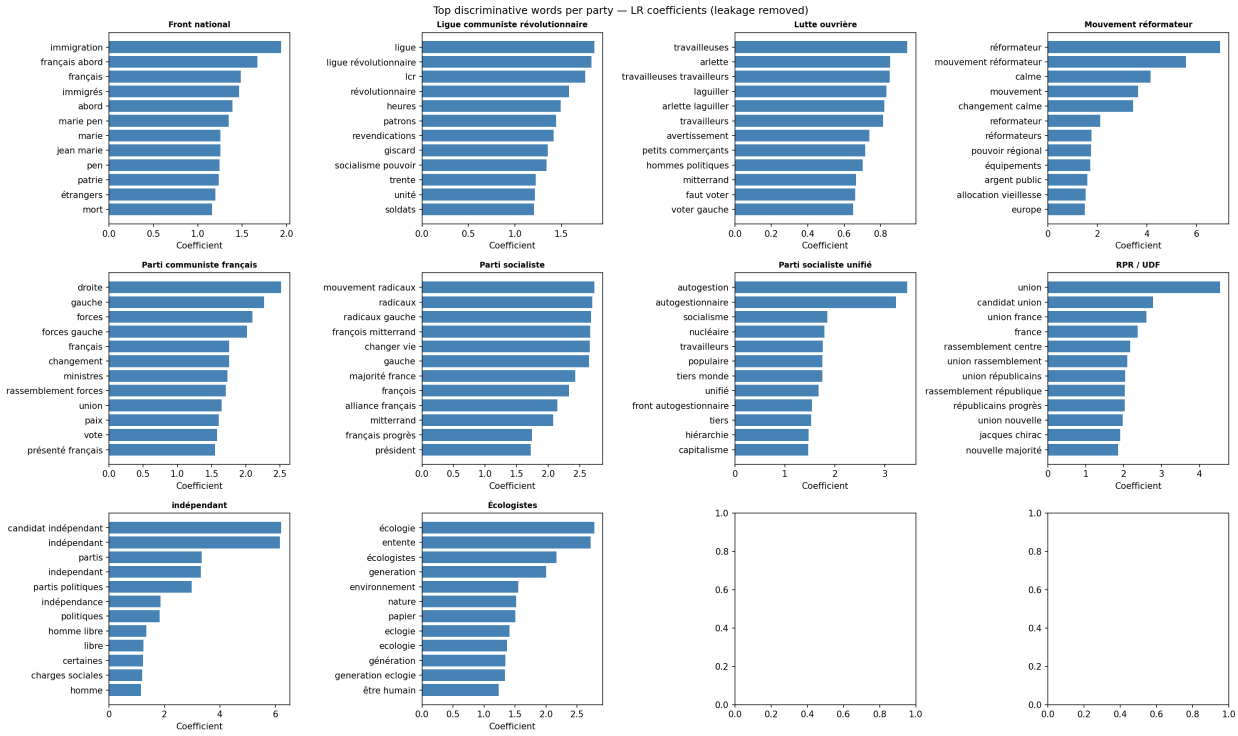


Figure 5: Top discriminative words per party (LR coefficients, leakage removed). Features are thematic and rhetorical, not party self-identifications.

register.

Importantly, this map was produced without any supervision on party positions. The algorithm had no access to party labels during LDA training.

## 4.4 Jensen-Shannon Divergence and Cross-Check with RQ1

Cosine similarity between party centroids is not well-suited for comparing probability distributions like LDA outputs. I instead use the Jensen-Shannon divergence (JSD), a symmetrised and bounded version of KL divergence:

$$\mathrm{JSD}(P\|Q) = \frac{1}{2}\,D_{\mathrm{KL}}(P\|M) + \frac{1}{2}\,D_{\mathrm{KL}}(Q\|M), \quad M = \tfrac{P+Q}{2}, \tag{1}$$

where JSD values range from 0 (identical distributions) to 1 (disjoint support).

Table 4 presents the JSD between the six most confused party pairs alongside the number of SVM classification errors from RQ1. The two analyses were conducted independently: LDA is unsupervised on topics, SVM is supervised on labels, and neither has access to the other's output. The alignment is therefore not a methodological artefact but reflects genuine political structure. Pairs with the lowest JSD generate the most errors.

The LCR/PSU row is particularly revealing. These two parties share a very similar topic distribution (JSD = 0.229), both dominated by labour and class struggle discourse (Topics 2 and 3), yet the SVM almost never confuses them. This points to a fundamental distinction between two notions of proximity: thematic proximity (captured by LDA, asking what themes a party discusses) and lexical distinctiveness (captured by TF-IDF, asking which specific words it uses to discuss them). LCR and PSU discuss the same themes using systematically different vocabularies, a distinction rooted in the organisational and rhetorical differences

Table 3: LDA topics: labels assigned after inspecting top words.

| Topic | Label | Key top words |
|---|---|---|
| 0 | Nationalist and immigration | français, immigration, immigrés, votez, impôts |
| 1 | General left/right discourse | gauche, droite, programme, changement, vie |
| 2 | Labour and class struggle | travailleurs, patrons, entreprises, emplois, salaire |
| 3 | Left electoral appeal | mitterrand, marchais, voter, gauche, contre |
| 4 | Electoral programme (general) | notre, france, majorité, candidat, politique |
| 5 | Ecology and nature | nature, animaux, écologistes, environnement, papier |
| 6 | Nationalist electoral | nationale, vote, avenir, force, populaire, suppression |
| 7 | Local and institutional | circonscription, suppléant, progrès, candidats, état |

Table 4: Cross-check: rhetorical proximity (JSD, lower is closer, range 0,1) vs. SVM misclassifications on the test set. The two analyses are fully independent.

| Party A | Party B | JSD (lower = closer) | SVM errors |
|---|---|---|---|
| Parti socialiste | RPR / UDF | 0.163 | 11 |
| Parti communiste français | Parti socialiste | 0.269 | 8 |
| indépendant | RPR / UDF | 0.106 | 7 |
| Front national | RPR / UDF | — | 4 |
| Mouvement réformateur | RPR / UDF | 0.123 | 3 |
| Parti socialiste unifié | Parti socialiste | — | 2 |
| LCR | Parti socialiste unifié | 0.229 | $\approx 0$ |

between Trotskyist and broader socialist traditions of the French radical left. Neither method alone could have established this.

It is also worth noting that the correlation between JSD and SVM errors is real but not perfect. The PS/RPR-UDF pair generates the most errors (11) with a JSD of 0.163, while the indépendant/RPR-UDF pair generates 7 errors with a lower JSD of 0.106. Part of this discrepancy is explained by class size: PS has 2,182 documents against 133 for indépendant, so more PS documents are exposed to potential confusion in absolute terms. This suggests that rhetorical proximity is the primary driver of misclassification, but class imbalance acts as a secondary amplifier.

# 5 RQ3: Temporal Stability of Political Vocabulary

## 5.1 Motivation

The random 80/20 split used in RQ1 draws training and test documents from the same joint distribution across all years (1973 to 1993). In practice, a political text classifier would be trained on past elections and deployed on future ones. RQ3 asks whether the lexical signatures learned from 1973 to 1981 manifestos are stable enough to generalise to 1988 and 1993, a genuine out-of-distribution evaluation that the random split cannot provide.

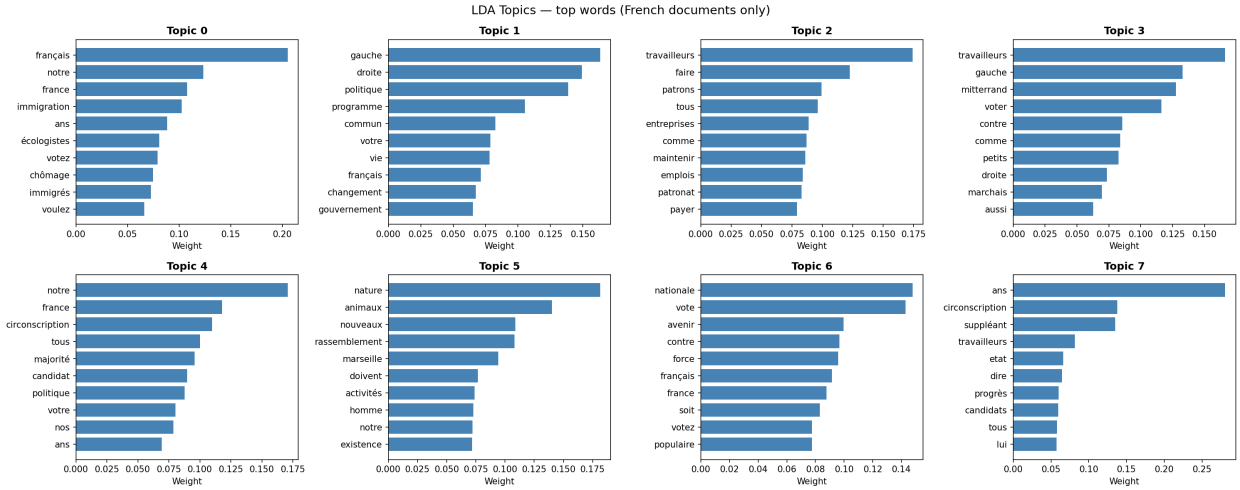


Figure 6: LDA topic top words (French documents only, Alsatian/German filtered). Topics are interpretable as distinct political thematic clusters.

## 5.2 Direct Temporal Split

I train the SVM on all documents from 1973, 1978, and 1981, and test on all documents from 1988 and 1993. To avoid additional confounding from the restructured party landscape, I retain only classes with at least 30 documents in both periods. Three classes are excluded: Écologistes (absent before 1988), PSU (1 document in 1993 only), and Mouvement réformateur (absent after 1973). The retained 7 parties yield 5,478 training and 4,371 test documents.

The SVM achieves F1 = 0.507 on this temporal split, compared to 0.963 on the random split, a drop of 0.456 F1 points. This degradation reflects two confounded factors.

**Factor 1: Genuine vocabulary drift.** Political language evolves over 20 years. New concepts emerge (cohabitation, European integration), old ones fade, and the rhetorical register of each party shifts as political conditions change.

**Factor 2: Class distribution shift.** Table 5 shows the dramatic change in candidate counts per party between the two periods. The FN grew by a factor of 6.7 while the PS shrank by 3.3 and LO by 3.4. A classifier trained on a PS-dominant distribution is tested on an FN-dominant one, which would degrade performance even if the vocabulary were perfectly stable.

Table 5: Class distribution shift between training (1973,1981) and test (1988,1993) periods. The FN and PS changes are major distributional confounders.

| **Party** | **Train** | **Test** | **Change** |
|---|---|---|---|
| Front national | 177 | 1,189 | $\times$ 6.7 |
| RPR / UDF | 1,213 | 1,239 | stable |
| Parti communiste français | 1,407 | 1,067 | $\div$ 1.3 |
| Parti socialiste | 1,671 | 511 | $\div$ 3.3 |
| Lutte ouvrière | 789 | 234 | $\div$ 3.4 |
| Ligue communiste révolutionnaire | 154 | 65 | $\div$ 2.4 |
| indépendant | 67 | 66 | stable |

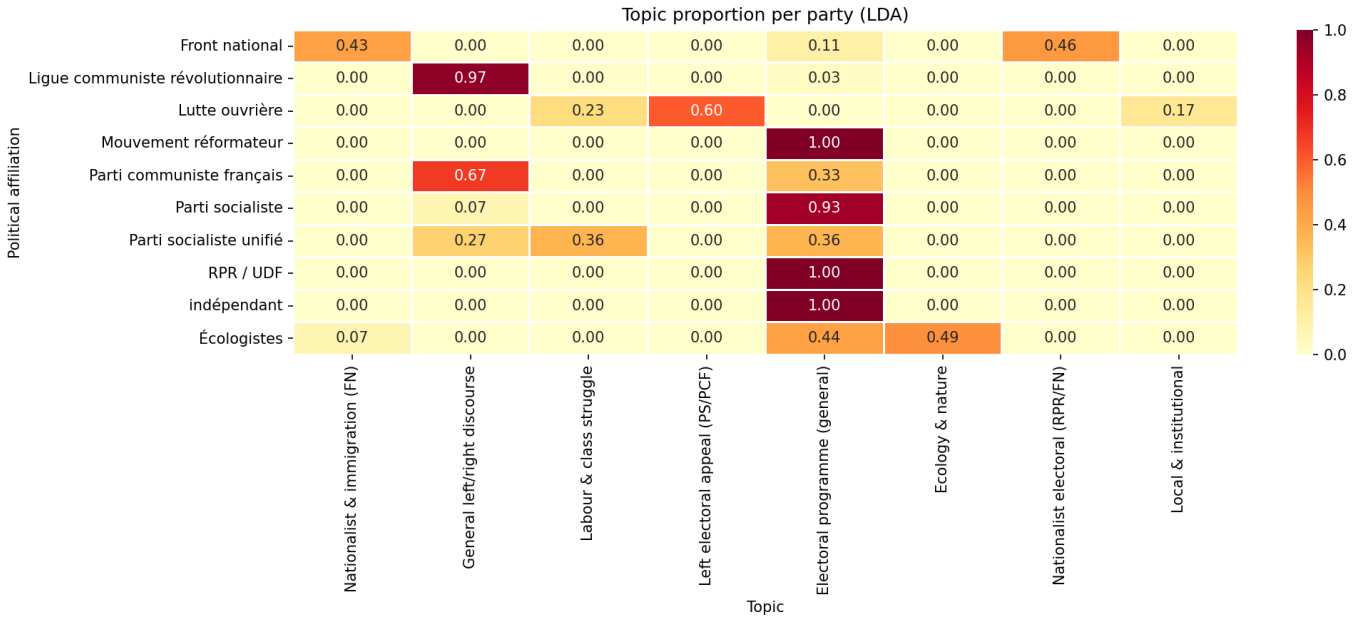


Figure 7: Topic proportion per party (LDA, French documents only). Each row sums to 1. The heatmap shows clear thematic clustering by political bloc.

### 5.3 Leave-One-Year-Out Evaluation

To better isolate genuine vocabulary drift from the distribution shift, I adopt a leave-one-year-out (LOO) protocol: for each of the five election years, I train on all documents from the other four years and evaluate on the held-out year, retaining only classes present in both train and test. This protocol is more balanced than the direct temporal split because each held-out year is surrounded by other years in the training set.

Results are shown in Figure 9 and Table 6.

Table 6: Leave-one-year-out F1 macro (TF-IDF + SVM). The random split baseline is shown for reference.

| **Held-out year** | **F1 macro** | **Gap vs. random split** |
|---|---|---|
| 1973 | 0.707 | −0.256 |
| 1978 | 0.620 | −0.343 |
| 1981 | 0.793 | −0.170 |
| 1988 | 0.762 | −0.201 |
| 1993 | 0.528 | −0.435 |
| Random split | 0.963 | reference |

The pattern is politically interpretable. 1981 achieves the highest LOO F1 (0.793): this was the peak of the classical left/right bipolarisation, with two dominant blocs (PS-PCF on the left, RPR-UDF on the right) that had well-defined and stable vocabularies. 1988 is the second easiest (0.762): the cohabitation period reinforced the bipolar structure despite some FN growth. 1978 is the hardest pre-1988 year (0.620): the right was still fragmented between Gaullists and Giscardiens before the RPR/UDF coalition consolidated, and the left had not yet fully converged around the PS. 1993 is the hardest year overall (0.528, gap of 0.435 vs. random split): a historic PS defeat, the FN surge to 12.4%, the first mass entry of the Écologistes, and several new centrist formations all simultaneously shift the vocabulary distribution away from every training

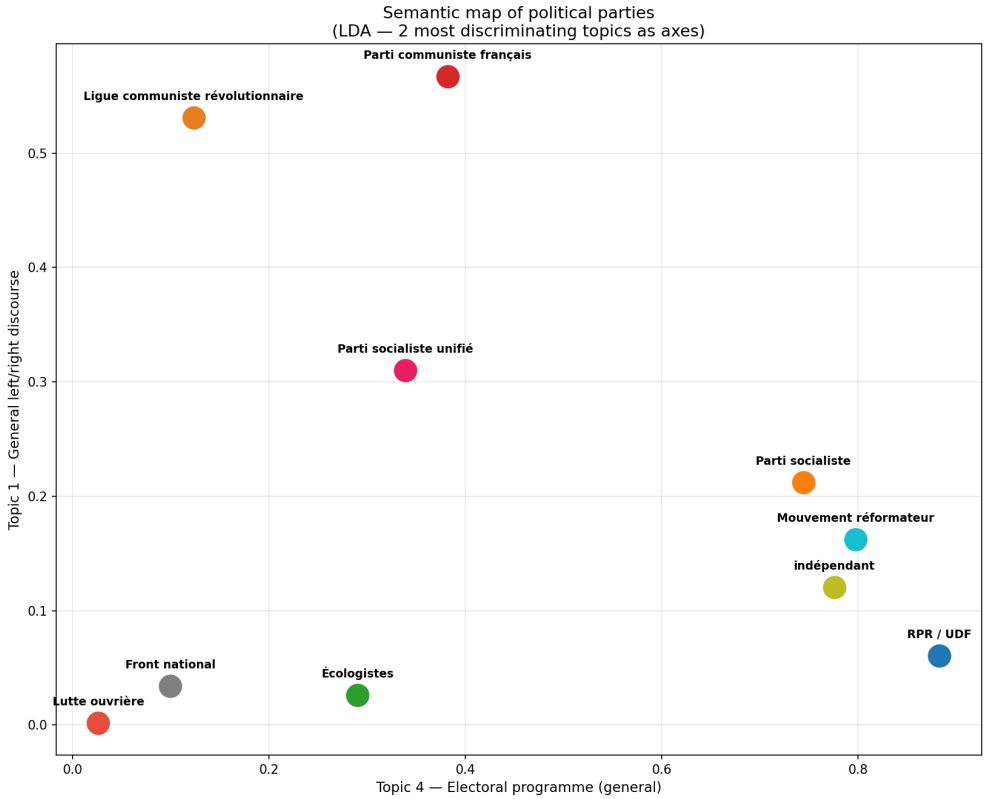


Figure 8: Semantic map of political parties in LDA topic space. Axes are the two topics with the highest variance across party centroids. Colours: dark red = PCF; orange = LCR; light red = LO; pink = PSU; orange-yellow = PS; green = Écologistes; cyan = Mouvement réformateur; blue = RPR/UDF; grey = FN; yellow = indépendant.

year. The year 1993 represents a genuine political rupture.

The gap between LOO F1 and the random split baseline (ranging from 0.170 to 0.435 depending on the year) quantifies the cross-temporal generalisation cost of each election. A stable political year (1981) incurs a moderate cost; a disrupted one (1993) incurs a large one. This suggests that LOO classifier performance could serve as a data-driven indicator of political discontinuity.

## 6 Discussion of the results

**Why do sparse TF-IDF models outperform dense embeddings?** The dominance of TF-IDF over FastText and LSI on this task can be explained by the structure of political language. French parties develop highly distinctive, repetitive lexical signatures: specific collocations like ”travailleurs travailleuses”, ”programme commun”, or ”français d’abord” that serve as ideological markers and are used consistently across all candidates of a given party. TF-IDF’s inverted document frequency weighting naturally amplifies exactly these party-specific terms while down-weighting generic vocabulary shared across parties. Dense methods like FastText and LSI compress this discriminative sparsity into a continuous representation, losing some of the sharp lexical boundaries between parties. For short political texts where party affiliation is expressed through a small set of recurring markers, this compression is more costly than helpful.

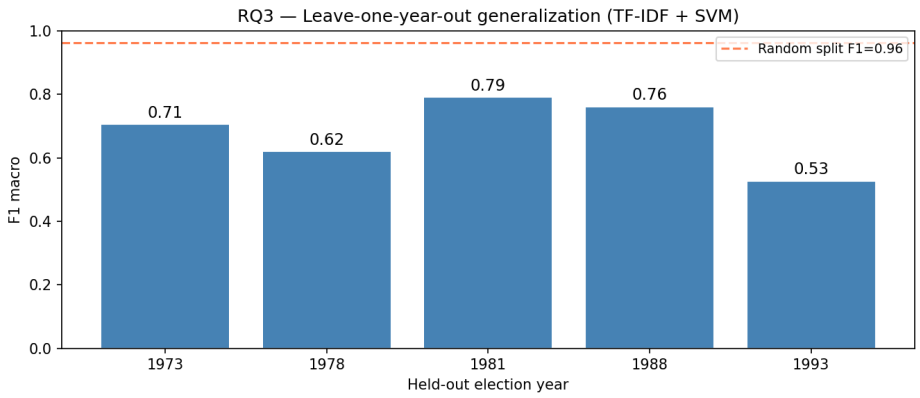


Figure 9: RQ3: leave-one-year-out F1 macro (TF-IDF + SVM). The dashed line marks the random split baseline (F1 = 0.963). Year-by-year variation tracks historical political disruption.

**Joint validation of RQ1 and RQ2.** The alignment between topic-space proximity (RQ2) and classification confusion (RQ1) is the central contribution of this paper. Its significance rests on the methodological independence of the two analyses: LDA is an unsupervised generative model trained on word co-occurrences with no access to party labels, while TF-IDF + SVM is a supervised discriminative model trained on party labels with no access to topic structure. Any convergence between their outputs must reflect genuine political structure in the text rather than a methodological artefact. The result confirms that the SVM's residual errors are not random or due to model limitations, but are structurally explained by the underlying rhetorical proximity of the parties.

**Thematic proximity vs. lexical distinctiveness.** The LCR/PSU case illustrates a fundamental distinction that the joint analysis uniquely enables. Both parties discuss the same themes (labour struggle, class conflict, left electoral politics) with similar proportions, giving a low JSD of 0.229 in topic space. Yet the classifier almost never confuses them, because they use systematically different vocabularies to discuss these shared themes: the LCR uses Trotskyist terminology ("révolutionnaire", "patronat", "arlette laguiller") while the PSU uses a broader socialist register ("autogestion", "programme", "tiers monde"). LDA captures what a party discusses; TF-IDF captures how it discusses it. This is something neither method could have revealed on its own.

**Temporal instability as a political signal.** The degradation from F1 = 0.963 to F1 = 0.507 is not a failure of the method but a discovery about the object of study. Political vocabulary is not stable across decades: the lexical signatures that parties develop in one period do not fully carry over to the next as the political landscape restructures. The year-by-year variation in LOO performance, higher in stable years like 1981 and 1988 and lower in disrupted ones like 1978 and 1993, provides a quantitative correlate of historical political change.

**Limitations.** Several limitations are worth noting. First, OCR noise introduces character-level errors in some documents, particularly for older scans. Second, the indépendant class aggregates genuinely independent candidates who share no common political project, making its low F1 a feature of the data rather than a model failure. Third, FastText is trained on a small corpus, which limits embedding quality. Fourth, the RPR/UDF label consolidation groups parties that did not always present a unified programme, particularly in 1973,1978.

# 7 Conclusion

In thiss study, I studied the Archelec corpus of French electoral manifestos (1973,1993) through classification, semantic mapping, and temporal evaluation, and I draw three main conclusions.

On **RQ1**, political affiliation can be predicted with high accuracy (F1 = 0.963, TF-IDF + Linear SVM) from leakage-free manifesto text. Sparse TF-IDF representations dominate all dense alternatives because French party rhetoric relies on distinctive, repetitive lexical signatures that the inverted document frequency weighting naturally amplifies. The leakage removal protocol is essential: without it, a classifier could achieve high accuracy by simply detecting party names, but the learned representations would have no real interpretability.

On **RQ2**, unsupervised LDA topic modelling reproduces the classical left/right axis of French politics without any supervision on party positions. The semantic map places left-wing radical parties in the upper-left quadrant, RPR/UDF in the lower-right, and the PS in the middle-right. The Jensen-Shannon divergence analysis reveals that parties close in topic space correspond exactly to the main confusion pairs of the classifier. This joint validation from two independent methods confirms the reality of the rhetorical structure in the data, and the the LCR/PSU case shows that thematic proximity and lexical distinctiveness are two separate dimensions that the combination of LDA and TF-IDF uniquely disentangles.

On **RQ3**, temporal generalisation degrades substantially (F1 = 0.507 on the direct temporal split, 0.528,0.793 leave-one-year-out), and the degradation is structured: 1993, the most politically disrupted election, is the hardest to predict, while 1981, the peak of bipolarisation, is the easiest. This finding inverts the usual framing of temporal degradation as a model failure: the classifier's cross-temporal performance serves as a quantitative indicator of political discontinuity.

Future work could improve FastText results with pre-trained French embeddings such as `cc.fr.300` or CamemBERT martin2020. A resampling approach that matches the class distribution between training and test periods would help disentangle vocabulary drift from distribution shift. Finally, incorporating candidate metadata (département, profession, electoral round) could further improve cross-temporal generalisation.

# References


[Bojanowski et al.2017] Bojanowski, P., Grave, E., Joulin, A., and Mikolov, T. (2017). Enriching word vectors with subword information. Transactions of the ACL, 5, 135,146.

[Martin et al.2020] Martin, L., et al. (2020). CamemBERT: a Tasty French Language Model. Proceedings of ACL 2020, 7203,7219.

[Sciences Po2020] Sciences Po / CEVIPOF (2020). Archelec: Archives électorales de Sciences Po. `https://archelec.sciencespo.fr`.